

Experiment 07:

Laboratory Exercise

1. Verification of The Sleuth Kit installation:

```
(student@kali)-[~/Shivam]
$ mmls -V
The Sleuth Kit ver 4.14.0
```

Observation:

The output confirms The Sleuth Kit version 4.14.0 is correctly installed and accessible from the command line. This validates that all TSK utilities (mmls, fsstat, fls, istat, icat, blkls) used later in the experiment are available.

2. Disk image selected for analysis:

2.1 Source disk identification

```
(student@kali)-[~/Shivam]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0  100G  0 disk
├─sda1       8:1    0   96G  0 part /
├─sda2       8:2    0    1K  0 part
├─sda5       8:5    0    4G  0 part [SWAP]
└─sr0       11:0    1 1024M  0 rom
```

Observation:

lsblk lists the source disk sda as a 100G disk with an ext4 root partition (sda1), an extended partition (sda2) and a 4G swap partition (sda5). This confirms the correct source device was identified before imaging.

2.2 Evidence directory creation

```
(student@kali)-[~/Shivam]
$ mkdir -p evidence
```

Observation:

A dedicated 'evidence' directory was created to store the acquired image, keeping forensic artefacts isolated from the working directory. This follows standard chain-of-custody practice.

2.3 Disk imaging using dd

```
(student@kali)-[~/Shivam]
└─$ sudo dd if=/dev/sda of=evidence/data.dd bs=4M status=progress
81952505856 bytes (82 GB, 76 GiB) copied, 97 s, 845 MB/s
dd: error writing 'evidence/data.dd': No space left on device
19616+0 records in
19615+0 records out
82273370112 bytes (82 GB, 77 GiB) copied, 97.5018 s, 844 MB/s
```

Observation:

The dd command copied approximately 82 GB (76.6 GiB) before terminating with a 'No space left on device' error. This indicates the destination volume ran out of storage during acquisition, producing a partial/truncated image rather than a complete bit-for-bit copy.

2.4 Image file verification

```
(student@kali)-[~/Shivam]
└─$ ls -lh evidence/data.dd
-rw-r--r-- 1 root root 77G Sep 24 11:33 evidence/data.dd
```

Observation:

The listing confirms the image file evidence/data.dd was created with a size of approximately 77 GB, consistent with the amount of data dd reported copying before the space error.

2.5 Hash verification

```
(student@kali)-[~/Shivam]
└─$ sha256sum evidence/data.dd
9169e8ce8b4b60410c6a50a45644b79dac8e1d8f0eeda931a55600b53c65c95a  evidence/data.dd
```

Observation:

A SHA-256 hash(9169e8ce8b4b60410c6a50a45644b79dac8e1d8f0eeda931a55600b53c65c-95a) was generated for the image file. This hash forms the baseline for integrity verification of the evidence throughout the investigation.

2.6 File type identification

```
(student@kali)-[~/Shivam]
└─$ file evidence/data.dd
evidence/data.dd: DOS/MBR boot sector
```

Observation:

The file command identifies the image as containing a DOS/MBR boot sector, confirming the acquired image begins at the disk's master boot record and is structurally a valid raw disk image.

2.7 Boot sector hex dump

```
(student@kali)-[~/Shivam]
$ xxd -l 512 evidence/data.dd
00000000: eb63 9010 8ed0 bc00 b0b8 0000 8ed8 8ec0 .c.....
00000010: fbbe 007c bf00 06b9 0002 f3a4 ea21 0600 ...|.....!..
00000020: 00be be07 3804 750b 83c6 1081 fefe 0775 ....8.u.....u
00000030: f3eb 16b4 02b0 01bb 007c b280 8a74 018b .....|...t..
00000040: 4c02 cd13 ea00 7c00 00eb fe00 0000 0000 L....|.....
00000050: 0000 0000 0000 0000 0000 0080 0100 0000 .....
00000060: 0000 0000 fffa 9090 f6c2 8074 05f6 c270 .....t...p
00000070: 7402 b280 ea79 7c00 0031 c08e d88e d0bc t...y|..1....
00000080: 0020 fba0 647c 3cff 7402 88c2 52be 807d . .d|<.t...R..}
00000090: e817 01be 057c b441 bbaa 55cd 135a 5272 .....|.A..U..ZRr
000000a0: 3d81 fb55 aa75 3783 e101 7432 31c0 8944 =..U.u7...t21..D
000000b0: 0440 8844 ff89 4402 c704 1800 668b 1e5c .@.D..D....f..\
000000c0: 7c66 895c 0866 8b1e 607c 6689 5c0c c744 |f.\.f..`|f.\..D
000000d0: 0600 70b4 42cd 1372 05bb 0070 eb76 b408 .p.B..r...p.v..
000000e0: cd13 730d 5a84 d20f 83d8 00be 8b7d e982 .s.Z.....}..
000000f0: 0066 0fb6 c688 64ff 4066 8944 040f b6d1 .f....d.@f.D....
00000100: c1e2 0288 e888 f440 8944 080f b6c2 c0e8 .....@.D.....
00000110: 0266 8904 66a1 607c 6609 c075 4e66 a15c .f..f..`|f..uNf.\
00000120: 7c66 31d2 66f7 3488 d131 d266 f774 043b |f1.f.4..1.f.t.;
00000130: 4408 7d37 fec1 88c5 30c0 c1e8 0208 c188 D.}7...0.....
00000140: d05a 88c6 bb00 708e c331 dbb8 0102 cd13 .Z...p..1.....
00000150: 721e 8cc3 601e b900 018e db31 f6bf 0080 r...`.....1...
00000160: 8ec6 fcf3 a51f 61ff 265a 7cbe 867d eb03 .....a.6Z|..}..
00000170: be95 7de8 3400 be9a 7de8 2e00 cd18 ebfe ..}.4...}.....
00000180: 4752 5542 2000 4765 6f6d 0048 6172 6420 GRUB .Geom.Hard
00000190: 4469 736b 0052 6561 6400 2045 7272 6f72 Disk.Read. Error
000001a0: 0d0a 00bb 0100 b40e cd10 ac3c 0075 f4c3 .....<.u..
000001b0: 0000 0000 0000 0000 b0e3 45d0 0000 8004 .....E.....
000001c0: 0104 83fe c2ff 0008 0000 0000 000c 00fe .....
000001d0: c2ff 0ffe c2ff fe0f 000c 02e8 7f00 0000 .....
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001f0: 0000 0000 0000 0000 0000 0000 0000 55aa .....U.
```

```
(student@kali)-[~/Shivam]
$ xxd -l 512 evidence/data.dd |tail
00000160: 8ec6 fcf3 a51f 61ff 265a 7cbe 867d eb03 .....a.6Z|..}..
00000170: be95 7de8 3400 be9a 7de8 2e00 cd18 ebfe ..}.4...}.....
00000180: 4752 5542 2000 4765 6f6d 0048 6172 6420 GRUB .Geom.Hard
00000190: 4469 736b 0052 6561 6400 2045 7272 6f72 Disk.Read. Error
000001a0: 0d0a 00bb 0100 b40e cd10 ac3c 0075 f4c3 .....<.u..
000001b0: 0000 0000 0000 0000 b0e3 45d0 0000 8004 .....E.....
000001c0: 0104 83fe c2ff 0008 0000 0000 000c 00fe .....
000001d0: c2ff 0ffe c2ff fe0f 000c 02e8 7f00 0000 .....
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001f0: 0000 0000 0000 0000 0000 0000 0000 55aa .....U.
```

Observation:

The hex dump of the first 512 bytes shows the MBR boot code, partition table entries and the 0x55AA boot-signature at the end of the sector, confirming the MBR structure is intact at the start of the image. The tail portion of the same MBR hex dump, isolating the partition table region and the closing 55AA signature bytes for closer inspection.

3. Partition analysis using fdisk:

```
(student@kali)-[~/Shivam]
$ fdisk -l evidence/data.dd
Failed to read extended partition table (offset=201330686): Success
Disk evidence/data.dd: 76.62 GiB, 82273370112 bytes, 160690176 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd045e3b0

Device            Boot    Start        End    Sectors  Size Id Type
evidence/data.dd1 *      2048  201328639  201326592   96G 83 Linux
evidence/data.dd2          201330686  209713151   8382466    4G  f W95 Ext'd (LBA)
```

Observation:

fdisk reports a DOS disklabel with a 96 GB Linux partition (data.dd1) and a 4 GB extended (W95 Ext'd/LBA) partition (data.dd2). A warning 'Failed to read extended partition table' at offset 201330686 confirms the image is missing data beyond that offset, consistent with the truncated dd acquisition.

4. File system information using fsstat:

```
(student@kali)-[~/Shivam]
$ fsstat -o 2048 evidence/data.dd
FILE SYSTEM INFORMATION
-----
File System Type: Ext4
Volume Name:
Volume ID: 7ce2427638108aa56a40e93ee54fa41d

Last Written at: 2026-09-24 11:14:38 (IST)
Last Checked at: 2026-09-23 15:52:02 (IST)

Last Mounted at: 2026-09-23 16:18:59 (IST)
Unmounted properly
Last mounted on: /

Source OS: Linux
Dynamic Structure
Compat Features: Journal, Ext Attributes, Resize Inode, Dir Index
InCompat Features: Filetype, Extents, 64bit, Flexible Block Groups,
Read Only Compat Features: Sparse Super, Large File, Huge File, Extra Inode Size

Journal ID: 00
Journal Inode: 8

METADATA INFORMATION
-----
Inode Range: 1 - 6291457
Root Directory: 2
Free Inodes: 5676585
Inode Size: 256

CONTENT INFORMATION
-----
Block Groups Per Flex Group: 16
Block Range: 0 - 25165823
Total Range in Image: 0 - 20086015
Block Size: 4096
```

```

Group: 766:
Block Group Flags: [INODE_UNINIT, BLOCK_UNINIT, INODE_ZEROED]
Inode Range: 6275073 - 6283264
Block Range: 25100288 - 25133055
Layout:
Data bitmap: 24641550 - 24641550
Inode bitmap: 24641566 - 24641566
Inode Table: 24648736 - 24649247
Data Blocks: 25100288 - 25133055
Free Inodes: 8192 (100%)
Free Blocks: 32768 (100%)
Total Directories: 0
Stored Checksum: 0x78A4

Group: 767:
Block Group Flags: [INODE_UNINIT, INODE_ZEROED]
Inode Range: 6283265 - 6291456
Block Range: 25133056 - 25165823
Layout:
Data bitmap: 24641551 - 24641551
Inode bitmap: 24641567 - 24641567
Inode Table: 24649248 - 24649759
Data Blocks: 25133056 - 25165823
Free Inodes: 8192 (100%)
Free Blocks: 32768 (100%)
Total Directories: 0
Stored Checksum: 0x1710

```

Observation:

fsstat identifies the file system as Ext4 with a root inode of 2, an inode range of 1–6,291,457 and 5,676,585 free inodes. The volume was last mounted on '/', unmounted properly, and shows journal, extents and 64-bit feature flags typical of a standard Linux root partition. The block group listing shows groups 766 and 767 fully unused (100% free inodes and blocks, zero directories), indicating these groups sit in the unallocated tail region of the file system that was not populated with data.

5. File and directory listing using fls:

5.1 Root directory listing

```

(student@kali)~[~/Shivam]
$ fls -o 2048 evidence/data.dd
d/- 5505025:  home
d/d 11:  lost+found
d/d 2883585:  etc
d/d 4718593:  media
l/l 17:  vmlinuz.old
d/d 131073:  var
d/d 3932161:  usr
d/- 5242881:  boot
d/d 3276801:  dev
d/d 3407873:  proc
d/- 5636097:  root
d/d 1966081:  run
d/d 524289:  sys
d/d 1572865:  tmp
l/l 13:  bin
l/l 14:  lib
l/l 15:  lib64
l/l 16:  sbin
d/d 262145:  mnt
d/d 2359297:  srv
d/d 1703937:  opt
l/l 21:  lib32
d/- 5767169:  .cache
l/l 18:  initrd.img.old
l/l 19:  vmlinuz
l/l 20:  initrd.img
V/V 6291457:  $OrphanFiles

```

Observation:

The root directory listing shows standard Linux top-level directories (home, etc, var, boot, usr, etc.) with their inode numbers, along with symbolic links such as vmlinuz and initrd.img, confirming a typical Debian/Kali root file system layout.

5.2 Identification of deleted directories

```
(student@kali)-[~/Shivam]
$ fls -o 2048 -r -d evidence/data.dd |grep '^d/-'
```

Observation:

This command recursively lists file system entries and filters for the 'd/-' marker, which flags directory entries whose metadata still exists but whose parent allocation has been removed (i.e. deleted directories).

5.3 Saving deleted directory entries

```
(student@kali)-[~/Shivam]
$ fls -o 2048 -r -d evidence/data.dd |grep '^d/-'> evidence/deleted_directories.txt
```

Observation:

The filtered results were saved to a text file (deleted_directories.txt) inside the evidence folder, preserving the list of deleted directory entries for later reference and reporting.

5.4 Recursive listing of deleted files

```
(student@kali)-[~/Shivam]
$ fls -o 2048 -r -d evidence/data.dd
-/- * 1(realloc):      etc/^
-/- * 1311543:  etc/ssl/certs/^^
-/- * 158:      var/lib/dpkg/info/d
-/- * 111:      var/lib/dpkg/info/4
-/- * 93:      var/lib/dpkg/info/^
-/- * 152:      var/lib/dpkg/info/7
-/- * 0:      var/lib/dpkg/info/$
-/- * 0:      var/lib/dpkg/info/
-/- * 0:      var/lib/dpkg/info/,
-/- * 0:      var/lib/dpkg/info/^
-/- * 2359636:  var/lib/dpkg/info/^^
r/r * 140955(realloc):  var/lib/systemd/deb-systemd-helper-enabled/.stateoh2yS.tmp
l/l * 134251(realloc):  var/lib/ghostscript/CMap/GBT-EUC-H.dpkg-new
l/l * 134253(realloc):  var/lib/ghostscript/CMap/GBT-EUC-V.dpkg-new
l/l * 134254(realloc):  var/lib/ghostscript/CMap/GBT-H.dpkg-new
l/l * 134265(realloc):  var/lib/ghostscript/CMap/GBpc-EUC-H.dpkg-new
l/l * 134268(realloc):  var/lib/ghostscript/CMap/GBpc-EUC-UCS2C.dpkg-new
l/l * 134275(realloc):  var/lib/ghostscript/CMap/HKdla-B5-H.dpkg-new
l/l * 134289(realloc):  var/lib/ghostscript/CMap/HKm314-B5-H.dpkg-new
l/l * 134298(realloc):  var/lib/ghostscript/CMap/HKscs-B5-H.dpkg-new
l/l * 134300(realloc):  var/lib/ghostscript/CMap/HKscs-B5-V.dpkg-new
l/l * 134305(realloc):  var/lib/ghostscript/CMap/Hiragana.dpkg-new
l/l * 134308(realloc):  var/lib/ghostscript/CMap/Hojo-EUC-V.dpkg-new
```

```

r/r * 4368082(realloc): usr/lib32/gconv/ISO8859-5.so.dpkg-new
r/r * 4368086(realloc): usr/lib32/gconv/ISO8859-9.so.dpkg-new
r/r * 4368094(realloc): usr/lib32/gconv/ISO_6937-2.so.dpkg-new
r/r * 4368095(realloc): usr/lib32/gconv/ISO_6937.so.dpkg-new
r/r * 4368096(realloc): usr/lib32/gconv/JOHAB.so.dpkg-new
r/r * 4368103(realloc): usr/lib32/gconv/LATIN-GREEK.so.dpkg-new
r/r * 4368105(realloc): usr/lib32/gconv/MAC-IS.so.dpkg-new
r/r * 4368108(realloc): usr/lib32/gconv/MACINTOSH.so.dpkg-new
r/r * 4368111(realloc): usr/lib32/gconv/NATS-SEFI.so.dpkg-new
r/r * 4368116(realloc): usr/lib32/gconv/SJIS.so.dpkg-new
r/r * 4368117(realloc): usr/lib32/gconv/T.61.so.dpkg-new
r/r * 4368121(realloc): usr/lib32/gconv/UHC.so.dpkg-new
r/r * 4368126(realloc): usr/lib32/gconv/VISCII.so.dpkg-new
r/r * 4368127(realloc): usr/lib32/gconv/gconv-modules.dpkg-new
r/r * 4368130(realloc): usr/lib32/gconv/libGB.so.dpkg-new
r/r * 4368132(realloc): usr/lib32/gconv/libJIS.so.dpkg-new
r/r * 4368133(realloc): usr/lib32/gconv/libJISX0213.so.dpkg-new
-/- * 0:      opt/microsoft/powershell/7/(
r/r * 1704506(realloc): opt/microsoft/powershell/7/System.Text.Encoding.dll.dpkg-new
r/r * 1704513(realloc): opt/microsoft/powershell/7/System.Reflection.Emit.dll.dpkg-new
r/r * 1704518(realloc): opt/microsoft/powershell/7/System.ValueTuple.dll.dpkg-new
r/r * 1704519(realloc): opt/microsoft/powershell/7/System.IO.FileSystem.Primitives.dll.dpkg-new
r/r * 1704520(realloc): opt/microsoft/powershell/7/System.Runtime.InteropServices.JavaScript.dll.dpkg-new
r/r * 1704526(realloc): opt/microsoft/powershell/7/System.Management.dll.dpkg-new
r/r * 1704535(realloc): opt/microsoft/powershell/7/System.Diagnostics.FileVersionInfo.dll.dpkg-new
r/r * 1704538(realloc): opt/microsoft/powershell/7/System.Security.Cryptography.ProtectedData.dll.dpkg-new
r/r * 1704542(realloc): opt/microsoft/powershell/7/System.Data.Common.dll.dpkg-new
r/r * 1704544(realloc): opt/microsoft/powershell/7/libSystem.IO.Compression.Native.so.dpkg-new
-/- * 0:      opt/microsoft/powershell/7/$
r/r * 1704593(realloc): opt/microsoft/powershell/7/System.Diagnostics.StackTrace.dll.dpkg-new
r/r * 1704599(realloc): opt/microsoft/powershell/7/System Buffers.dll.dpkg-new
r/r * 1704613(realloc): opt/microsoft/powershell/7/System.IO.UnmanagedMemoryStream.dll.dpkg-new
r/r * 1704614(realloc): opt/microsoft/powershell/7/System.Threading.Overlapped.dll.dpkg-new
-/r * 2884531(realloc): opt/microsoft/powershell/7/γ^
-/d * 2884555(realloc): opt/microsoft/powershell/7/♦^$

```

Observation:

Recursive listing with the `-d` flag reveals numerous deleted file entries (marked with an asterisk), largely from apt package cache and dpkg temporary files, showing routine package-manager housekeeping activity rather than user-initiated deletion. Further scrolled output shows additional deleted entries from library and PowerShell package directories, reinforcing that most recovered 'deleted' artefacts originate from software installation/removal rather than evidentiary user files.

6. Metadata analysis using `istat`:

```

(student@kali)-[~/Shivam]
$ istat -o 2048 evidence/data.dd 140955
inode: 140955
Allocated
Group: 17
Generation Id: 4037820621
uid / gid: 0 / 0
mode: rrw-r--r--
Flags: Extents,
size: 57
num of links: 1

Inode Times:
Accessed:      2026-09-23 15:59:31.275991653 (IST)
File Modified: 2026-09-23 15:59:31.282967327 (IST)
Inode Modified: 2026-09-23 15:59:31.282967327 (IST)
File Created:  2026-09-23 15:59:31.275991653 (IST)

Direct Blocks:

```

Observation:

With a valid inode (140955) supplied, istat returns full metadata: the inode is allocated, owned by uid/gid 0, is 57 bytes in size, and shows matching MACB (Modified / Accessed / Changed/ Born) timestamps of 2026-09-23 15:59:31 IST, indicating the file was created and immediately closed without later modification.

7. File recovery using icat:**7.1 Creation of recovery directory**

```
(student@kali)-[~/Shivam]
$ mkdir recovered_files
mkdir: cannot create directory 'recovered_files': No space left on device
```

Observation:

Before recovery could proceed, creating a destination folder for recovered files failed because the analysis workstation itself was out of disk space, an environmental issue separate from the evidence image.

7.2 File recovery using icat

```
(student@kali)-[~/Shivam]
$ icat -o 2048 evidence/data.dd 140955> recovered_report.txt
Error reading image file (tsk_fs_read_block: Address missing in partial image: 22196415) (tsk_fs_file_walk: Error reading block at 22196415)

(student@kali)-[~/Shivam]
$ icat -o 2048 evidence/data.dd 140955>recovered_report.txt
Error reading image file (tsk_fs_read_block: Address missing in partial image: 22196415) (tsk_fs_file_walk: Error reading block at 22196415)

(student@kali)-[~/Shivam]
$ icat -o 2048 evidence/data.dd 140955> recovered_report.txt
Error reading image file (tsk_fs_read_block: Address missing in partial image: 22196415) (tsk_fs_file_walk: Error reading block at 22196415)

(student@kali)-[~/Shivam]
$ icat -o 2048 evidence/data.dd 140955 > recovered_report.txt
Error reading image file (tsk_fs_read_block: Address missing in partial image: 22196415) (tsk_fs_file_walk: Error reading block at 22196415)
```

Observation:

All repeated attempts to extract inode 140955 with icat failed with 'Error reading image file... Address missing in partial image', because block 22196415 lies beyond the range actually captured by the truncated dd acquisition. This confirms the earlier dd space error left the image incomplete, preventing full recovery of that file's content.

7.3 Verification of recovered output file



Observation:

The output-redirection target recovered_report.txt was created in the home directory despite icat's failure, but contains no recovered data - it is only an artefact of the shell's '>' redirection, not a successful file carve.

8. Extraction of unallocated space using tsk_recover:

8.1 Creation of recovery directory

```
(student@kali)-[~/Shivam]
$ mkdir recover_deleted
mkdir: cannot create directory 'recover_deleted': No space left on device
```

Observation:

As with the earlier recovery attempt, creating a local output directory for carved files failed due to the workstation running out of disk space, blocking the extraction workflow before it could start.

8.2 Initial recovery command attempts

```
(student@kali)-[~/Shivam]
$ tsk_recover -q data.dd recovered_files
tsk_recover: invalid option -- 'q'
Invalid argument: data.dd
usage: tsk_recover [-vVae] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o sector_offset] [-P pooltype]
        -i imgtype: The format of the image file (use '-i list' for supported types)
        -b dev_sector_size: The size (in bytes) of the device sectors
        -f fstype: The file system type (use '-f list' for supported types)
        -v: verbose output to stderr
        -V: Print version
        -a: Recover allocated files only
        -e: Recover all files (allocated and unallocated)
        -o sector_offset: sector offset for a volume to recover (recovers only that volume)
        -P pooltype: Pool container type (use '-P list' for supported types)
        -B pool_volume_block: Starting block (for pool volumes only)
        -d dir_inum: Directory inum to recover from (must also specify a specific partition using -o o

(student@kali)-[~/Shivam]
$ tsk_recover -a data.dd recovered_files
Error stat(ing) image file (raw_open: image "data.dd" - No such file or directory)
```

Observation:

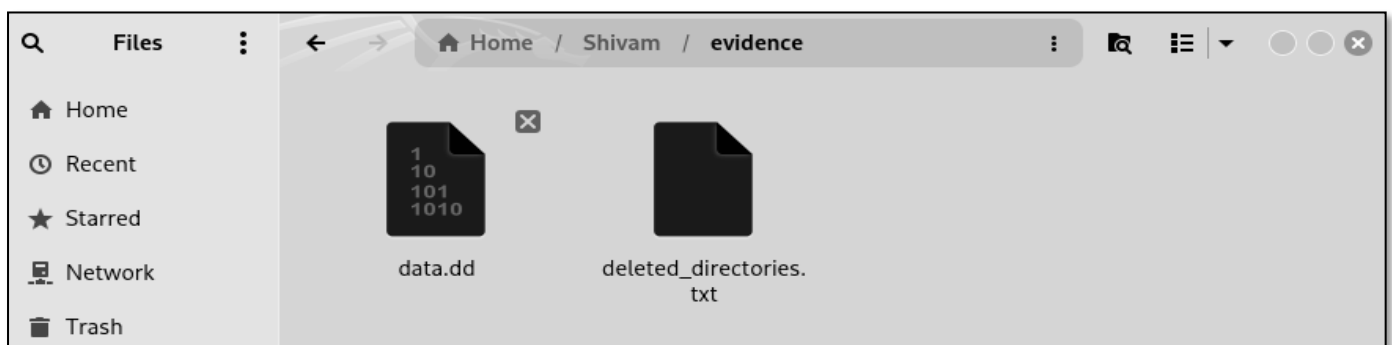
The first command used an unsupported '-q' option and was rejected; the corrected command then failed with 'No such file or directory' because it referenced data.dd directly instead of the correct path evidence/data.dd.

8.3 Final recovery attempt

```
(student@kali)-[~/Shivam]
└─$ tsk_recover -e data.dd recover_deleted
Error stat(ing) image file (raw_open: image "data.dd" - No such file or directory)
```

Observation:

A further attempt to recover all (allocated and unallocated) files again failed for the same path reason, meaning unallocated-space extraction was not successfully completed in this run and would need to be repeated with the correct evidence path and sufficient free disk space.

9. Observations from recovered evidence.**Observation:**

The evidence folder retains the acquired disk image (data.dd) and the extracted deleted_directories.txt listing as the two tangible outputs of this experiment, forming the basis for the summary observations below.